

1. Use the Linux terminal to create a new partition on a virtual disk using the **fdisk** tool, then format it as **ext4** and mount it to **/mnt/music**.

```
vboxuser@Meridian-Linux-Server:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda          8:0      0 50.5G  0 disk
├─sda1       8:1      0    1M  0 part
└─sda2       8:2      0 50.5G  0 part  /
sdb          8:16     0    5G   0 disk
├─md0        9:0      0    5G   0 raid1 /mnt/raid1
└─sdc        8:32     0    5G   0 disk
   └─sdc1     8:33     0    5G   0 part  /mnt/music
sr0         11:0     1 1024M  1 rom

vboxuser@Meridian-Linux-Server:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            680M  1.2M  679M   1% /run
/dev/sda2        50G   3.5G   44G   8% /
tmpfs            1.7G     0   1.7G   0% /dev/shm
tmpfs            1.7G     0   1.7G   0% /tmp
none            1.0M     0   1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0   1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0   1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M     0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs           340M    8.0K  340M   1% /run/user/1000
/dev/md0         4.9G   1.3M   4.6G   1% /mnt/raid1
/dev/sdc1        4.9G   1.3M   4.6G   1% /mnt/music

vboxuser@Meridian-Linux-Server:~$
```

2. Set up a Logical Volume Management (LVM) group named 'media_vg' with two physical volumes, then create a logical volume 'photos_lv' of 1GB and mount it to **/mnt/photos**.

```
vboxuser@Meridian-Linux-Server:~$ sudo pvcreate -y -ff /dev/sdb /dev/sdc
[sudo: authenticate] Password:
Physical volume "/dev/sdb" successfully created.
Physical volume "/dev/sdc" successfully created.
vboxuser@Meridian-Linux-Server:~$ sudo vgcreate media_vg /dev/sdb /dev/sdc
Volume group "media_vg" successfully created
vboxuser@Meridian-Linux-Server:~$ sudo lvcreate -L 1G -n photos_lv media_vg
WARNING: ext4 signature detected on /dev/media_vg/photos_lv at offset 1080. Wipe it? [y/n]: y
Wiping ext4 signature on /dev/media_vg/photos_lv.
Logical volume "photos_lv" created.
vboxuser@Meridian-Linux-Server:~$ sudo mkfs.ext4 /dev/media_vg/photos_lv
mke2fs 1.47.2 (1-Jan-2025)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: bf2674bf-8bdf-4533-9c65-ab2572d0b7ce
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

vboxuser@Meridian-Linux-Server:~$ sudo mkdir -p /mnt/photos
vboxuser@Meridian-Linux-Server:~$ sudo mount /dev/media_vg/photos_lv /mnt/photos
vboxuser@Meridian-Linux-Server:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE  MOUNTPOINTS
sda          8:0      0 50.5G  0 disk
├─sda1       8:1      0    1M  0 part
└─sda2       8:2      0 50.5G  0 part  /
sdb          8:16     0    5G   0 disk
├─media_vg-photos_lv 252:0     0    1G   0 lvm   /mnt/photos
└─sdc        8:32     0    5G   0 disk
sr0         11:0     1 1024M  1 rom

vboxuser@Meridian-Linux-Server:~$ sudo vgs
VG      #PV #LV #SN Attr   VSize VFree
media_vg  2  1  0 wz--n- 9.99g 8.99g

vboxuser@Meridian-Linux-Server:~$
```

3. Simulate a hardware failure by unmounting a mounted partition and deleting its mount directory, then use Linux commands to diagnose and recover access to the data.

```
vboxuser@Meridian-Linux-Server:~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0      0 50.5G  0 disk
├─sda1                              8:1      0    1M  0 part
└─sda2                              8:2      0 50.5G  0 part /
sdb                                  8:16     0    5G   0 disk
└─media_vg-photos_lv              252:0     0    1G   0 lvm  /mnt/photos
sdc                                  8:32     0    5G   0 disk
sr0                                  11:0     1 1024M  1 rom
vboxuser@Meridian-Linux-Server:~$ ls -la /mnt/photos
total 24
drwxr-xr-x 3 root root 4096 Sep 20 06:23 .
drwxr-xr-x 5 root root 4096 Sep 20 06:36 ..
drwx----- 2 root root 16384 Sep 20 06:23 lost+found
vboxuser@Meridian-Linux-Server:~$
```

4. Resize the logical volume 'photos_lv' you created earlier to increase its size by 500MB, then verify the new size using the appropriate LVM and filesystem commands.

```
vboxuser@Meridian-Linux-Server:~$ sudo lvextend -r -L +500M /dev/mapper/media_vg-photos_lv
File system ext4 found on media_vg/photos_lv mounted at /mnt/photos.
Size of logical volume media_vg/photos_lv changed from 1.00 GiB (256 extents) to <1.49 GiB (381 extents).
Extending file system ext4 to <1.49 GiB (1598029824 bytes) on media_vg/photos_lv...
resize2fs /dev/media_vg/photos_lv
resize2fs 1.47.2 (1-Jan-2025)
Filesystem at /dev/media_vg/photos_lv is mounted on /mnt/photos; on-line resizing required
old_desc_blocks = 1, new_desc_blocks = 1
The filesystem on /dev/media_vg/photos_lv is now 390144 (4k) blocks long.

resize2fs done
Extended file system ext4 on media_vg/photos_lv.
Logical volume media_vg/photos_lv successfully resized.
vboxuser@Meridian-Linux-Server:~$ sudo lvs media_vg/photos_lv
LV      VG      Attr      LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
photos_lv media_vg -wi-ao--- <1.49g
vboxuser@Meridian-Linux-Server:~$ lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0      0 50.5G  0 disk
├─sda1                              8:1      0    1M  0 part
└─sda2                              8:2      0 50.5G  0 part /
sdb                                  8:16     0    5G   0 disk
└─media_vg-photos_lv              252:0     0    1.5G  0 lvm  /mnt/photos
sdc                                  8:32     0    5G   0 disk
sr0                                  11:0     1 1024M  1 rom
vboxuser@Meridian-Linux-Server:~$ df -h /mnt/photos
Filesystem                Size  Used Avail Use% Mounted on
/dev/mapper/media_vg-photos_lv 1.5G 280K  1.4G   1% /mnt/photos
vboxuser@Meridian-Linux-Server:~$
```

5. A user reports that their USB drive is not detected when plugged in. List 3 Linux commands you would use to troubleshoot this issue and explain what each command checks.

1) lsusb

- Checks whether Linux detects the USB device at the hardware/USB level.
- It lists all connected USB devices.

2) dmesg | tail -20

- Checks the latest kernel messages for USB related errors or connection events.
- Useful for seeing whether Linux recognized the USB when it was plugged in.

3) lsblk

- Checks whether the USB drive appears as a storage device.
- It also shows partitions and their sizes.